NEXT GEN SOFTWARE HUB

CYBER SECURITY INTERNSHIP

WEEK 04 TASK 04 REPORT

Introduction to Security Tools

Exploring Essential Tools for Scanning, Exploitation and Web Security

TOPICS

- Nmap
- Netcat
- Nikto
- Burp Suite (Community)

TOOLS

- Nmap
- Netcat
- Nikto
- Burp Suite (Community)

RESOURCES

- Nmap Guide
- Burp Suite Guide

TASK

Run a port scan using Nmap and analyze output.

Week 4: Introduction to Security Tools

PLANS :

Phase 1: Foundations (Weeks 1–4)



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2026

1. INTRODUCTION

Cybersecurity professionals use a variety of security tools to identify, analyze, and assess systems and applications. These tools provide visibility into network services, web applications, open ports, configurations, and potential security weaknesses.

Security tools are particularly important during penetration testing and vulnerability assessment because they allow security professionals to collect information about an authorized target before performing further security analysis.

This task focuses on the introduction and practical use of common cybersecurity tools, including Nmap, Netcat, Nikto, and Burp Suite.

Nmap is primarily used for network discovery, port scanning, service identification, and service version detection. Netcat can be used to establish network connections and test whether particular services or ports are accessible. Nikto is a web server scanner that can identify potentially interesting files, configurations, outdated software, and common web server security issues. Burp Suite is a web application security testing platform that allows HTTP requests and responses to be intercepted and analyzed.

The practical work was performed in a controlled virtual laboratory using Kali Linux as the security testing machine and DC-1 from VulnHub as the intentionally vulnerable target.

The target was first discovered on the local network using Netdiscover. Nmap was then used to scan the target and identify open ports and services. Nikto was used to perform a basic web server assessment, while Burp Suite was configured as a proxy to intercept and inspect HTTP traffic between the browser and the target web application.

The practical demonstrated how different security tools can provide different types of information about the same target and how the information collected during reconnaissance can help security professionals understand the attack surface of a system.

2. OBJECTIVES

The objectives of this task were:

- To understand the purpose of common cybersecurity tools.
- To understand the purpose of Nmap.
- To understand the purpose of Netcat.
- To understand the purpose of Nikto.
- To understand the purpose of Burp Suite.
- To understand network reconnaissance.
- To identify an authorized target on a local network.
- To identify open ports on a target system.
- To identify services running on open ports.
- To identify service versions using Nmap.
- To perform a web server assessment using Nikto.
- To understand HTTP request interception using Burp Suite.
- To analyze information obtained from security tools.
- To understand how exposed services contribute to the attack surface.
- To connect theoretical security concepts with practical security testing.

3. INTRODUCTION TO SECURITY TOOLS

Security tools are software applications designed to assist cybersecurity professionals with tasks such as network discovery, vulnerability assessment, traffic analysis, penetration testing, monitoring, and security investigations.

Different tools provide different types of information.

For example:

Nmap can identify:

- Hosts
- Open ports
- Services
- Service versions
- Operating system information in some scan configurations

Netcat can be used to:

- Test TCP/UDP connectivity
- Connect to network services
- Listen for network connections
- Troubleshoot network communication

Nikto can identify:

- Web server information
- Potentially dangerous files
- Common web server misconfigurations
- Missing security headers
- Interesting web resources

Burp Suite can be used to:

- Intercept HTTP requests
- Inspect HTTP responses
- Modify requests in an authorized test environment
- Analyze cookies and headers
- Map web application functionality
- Perform web application security testing

Using multiple tools together can provide a broader view of a target than relying on a single tool.

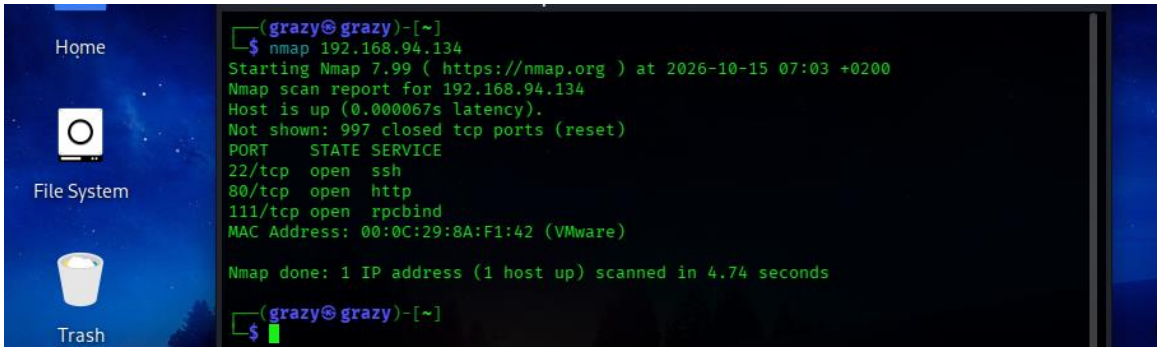
4. NMAP

Nmap stands for Network Mapper.

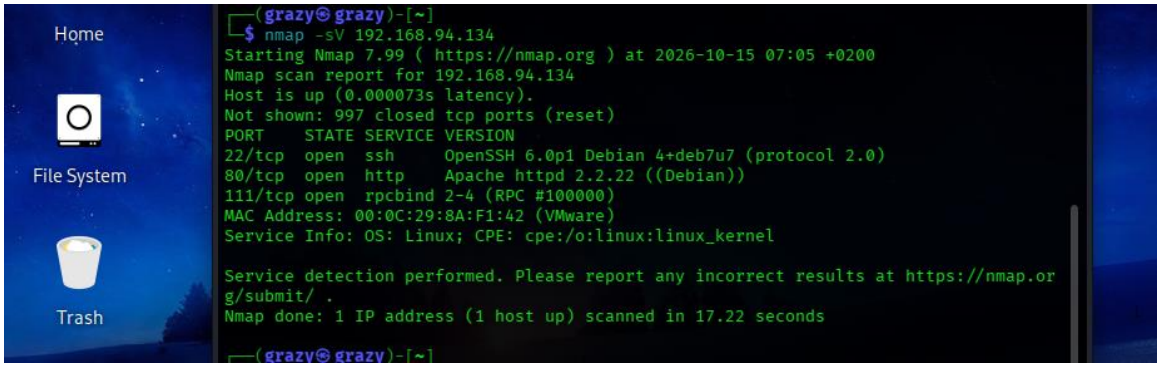
It is a network discovery and security auditing tool used to identify hosts, open ports, services, and service versions on authorized systems.

Nmap is commonly used during the reconnaissance stage of a penetration test.

A basic Nmap scan can identify common open TCP ports:



Service and version detection can provide additional information:



In this practical, a full TCP port scan with service detection was performed: **nmap -sV -p- 192.168.94.134**

The options used have the following meanings:

-sV: Attempts to identify the service and version running on discovered open ports.

-p-: Scans all TCP ports from 1 through 65535 rather than only Nmap's default port set.

Therefore, the command was useful for determining which TCP services were exposed by the DC-1 machine.

5. IMPORTANCE OF PORT SCANNING

A port represents a logical communication endpoint on a computer.

Network services commonly listen on specific ports.

Examples include:

Port	Common Service
22	SSH
53	DNS
80	HTTP
443	HTTPS

21 FTP

25 SMTP

Port scanning allows a security professional to determine which services are accessible on a target.

This information is important because an exposed service may increase the attack surface of a system.

For example, if a web server is running on port 80, the security professional can perform additional web application testing.

If SSH is running on port 22, the tester can assess the SSH configuration and determine whether the service is appropriately secured.

6. NETDISCOVER

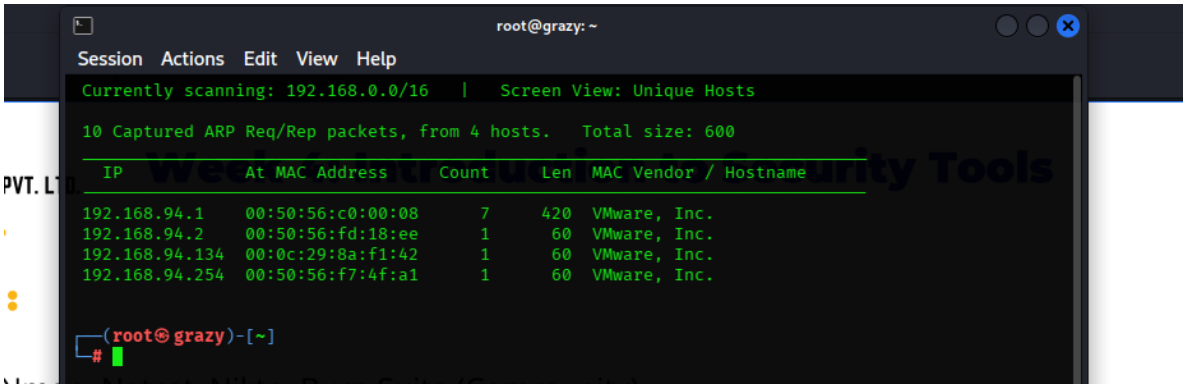
Netdiscover is a network discovery tool commonly used to identify hosts on a local network.

It uses ARP-based discovery to identify devices that are communicating on the local network.

In this practical, Netdiscover was used before Nmap to identify the IP address of the DC-1 target.

The target was identified as: **192.168.94.134**

The discovery process established the target IP address that was subsequently used during the Nmap scan.



The screenshot provides evidence that the target machine was discovered on the authorized virtual network before the scanning process began.

7. PRACTICAL ENVIRONMENT

The practical was performed in a controlled virtual laboratory.

Security testing machine

Operating System: Kali Linux

Target machine

System: DC-1 from VulnHub

Target IP address

192.168.94.134

Tools used

- Netdiscover
- Nmap
- Netcat
- Nikto
- Burp Suite
- Firefox

The target was an intentionally vulnerable virtual machine used for cybersecurity learning and testing.

8. PRACTICAL METHODOLOGY

The practical followed the following process:

1. Started Kali Linux.
2. Started the DC-1 virtual machine.
3. Used Netdiscover to identify devices on the local network.
4. Identified DC-1 at 192.168.94.134.
5. Used Nmap to scan the target.
6. Scanned all TCP ports using -p-.
7. Used service detection using -sV.
8. Identified the services running on the open ports.
9. Analyzed the Nmap results.
10. Performed a web server assessment using Nikto.
11. Identified web server and application information.
12. Opened the DC-1 web application using Firefox.
13. Configured Burp Suite as an HTTP proxy.
14. Intercepted HTTP traffic generated by the browser.
15. Examined HTTP headers, cookies, requests, and responses.
16. Used the collected information to understand the target's attack surface.

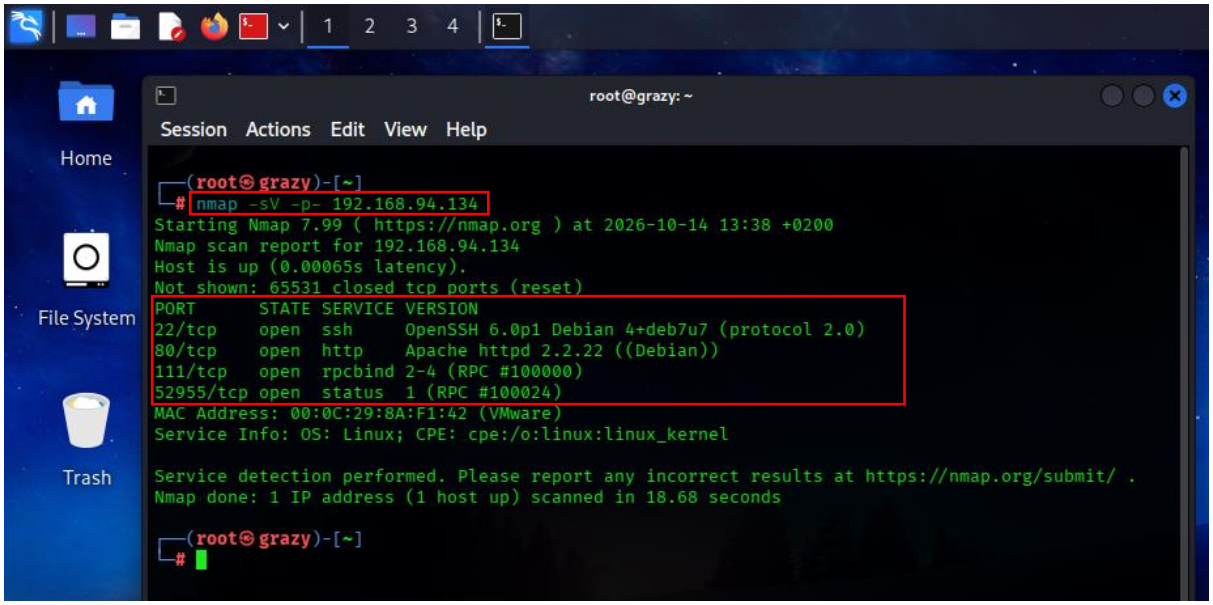
9. NMAP PORT SCANNING

After identifying the target IP address, Nmap was used to scan DC-1.

The command used was: **nmap -sV -p- 192.168.94.134**

The scan identified several open TCP ports.

The results included:



The screenshot provides direct evidence that Nmap was used to identify the network services exposed by the DC-1 machine and the scan therefore showed that the target was running multiple network services.

10. ANALYSIS OF NMAP RESULTS

The Nmap scan provides important information about the network attack surface of the DC-1 machine. Network services communicate through logical ports, and each open port may represent a different function provided by the target system.

The purpose of analysing Nmap results is not simply to identify which ports are open, but to understand what services are exposed, why those services may be required, and what security considerations are associated with them.

The scan identified four open TCP ports:

Port	Service	Version
22/tcp	SSH	OpenSSH 6.0p1 Debian 4+deb7u7
80/tcp	HTTP	Apache httpd 2.2.22 (Debian)
111/tcp	Rpcbind	2-4
52955/tcp	Status	1

The following subsections analyse each discovered service.

10.1 Port 22 - SSH

Port 22/tcp is associated with Secure Shell (SSH). SSH is a network protocol designed to provide secure remote access to computers over a network.

SSH is commonly used by system administrators to remotely manage Linux and Unix systems. Through an SSH connection, an authorized administrator can perform tasks such as managing files, configuring services, monitoring the system, and administering the operating system.

The Nmap scan identified the service as:

OpenSSH 6.0p1 Debian 4+deb7u7

This is significant because service-version detection provides information about the specific SSH implementation running on the target rather than simply reporting that port 22 is open.

Security significance

From a cybersecurity perspective, an exposed SSH service increases the remotely accessible attack surface of a system.

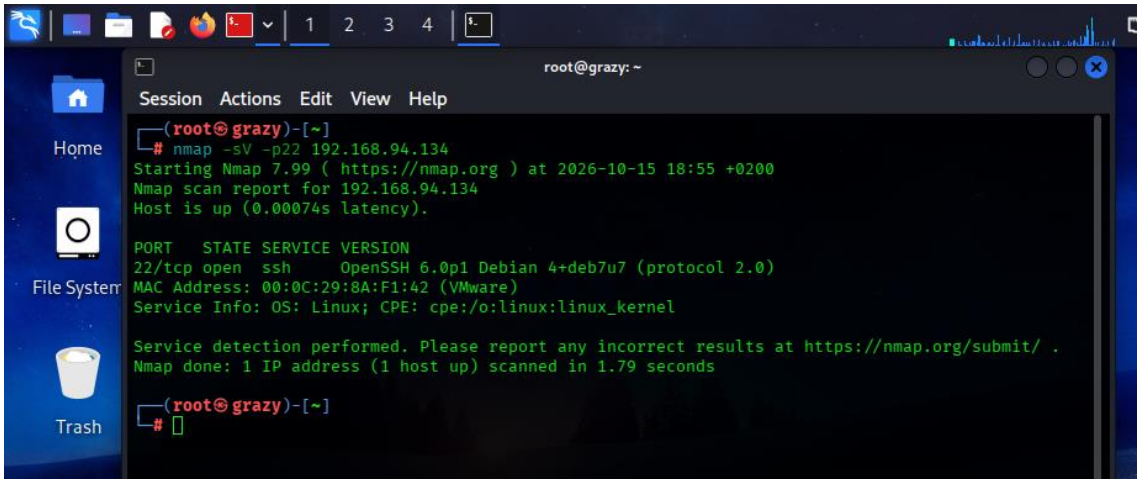
However, the presence of SSH does not automatically mean that the service is vulnerable. SSH may be legitimately required for remote administration. The security concern depends on factors such as its configuration, authentication controls, access restrictions, and whether the software is appropriately maintained.

During a security assessment, an analyst would therefore consider questions such as:

- Is remote SSH access required?
- Who is permitted to access the service?
- Is authentication properly configured?
- Are unnecessary accounts prevented from accessing SSH?
- Is access restricted to trusted networks?
- Is the installed SSH implementation appropriately maintained?

These questions demonstrate why service identification is important during reconnaissance.

Practical verification



The screenshot provides practical evidence that SSH is exposed on the target and that Nmap was able to identify the SSH implementation.

The key finding is therefore: **Port 22 is open and provides an SSH remote-access service running OpenSSH 6.0p1 Debian 4+deb7u7.**

10.2 Port 80 - HTTP

Port 80/tcp is traditionally associated with HTTP (Hypertext Transfer Protocol).

HTTP is one of the fundamental protocols used to deliver web pages and web applications between web servers and clients such as Firefox.

The Nmap scan identified: **Apache httpd 2.2.22 (Debian)**

This means that the DC-1 machine is running an Apache web server and is accepting HTTP connections.

Security significance

A web server represents an important part of a system's attack surface because it can provide access to web applications and resources.

Unlike SSH, which primarily provides remote administration functionality, HTTP provides an interface intended for interaction through a web browser.

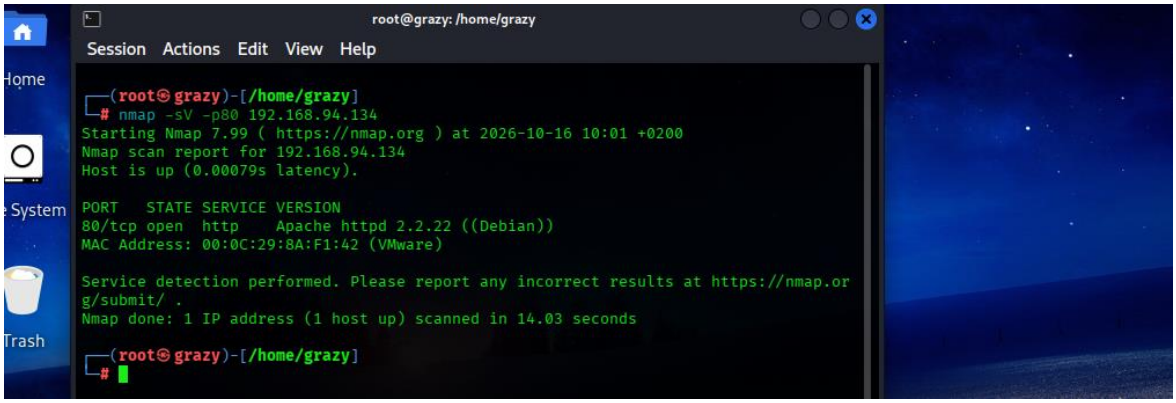
A web server may expose:

- Web pages
- Login forms
- Application functionality
- Directories
- Configuration information
- Cookies
- HTTP headers
- Application resources

Therefore, once Nmap identifies port 80, a security professional can move from network-level reconnaissance to web-server and web-application assessment.

This is exactly what happened in this practical. The discovery of port 80 led to the use of Nikto, Firefox, and later Burp Suite.

Practical verification



The result confirms that the target has an accessible HTTP service and identifies the web-server software.

The important finding is: **Port 80 is open and is running Apache httpd 2.2.22 on Debian, providing a web-based attack surface for further assessment**

10.3 Port 111 - RPCbind

Port 111/tcp was identified as rpcbind.

RPC stands for Remote Procedure Call. RPC allows applications to request services from another process, potentially running on another system.

rpcbind assists RPC clients in determining which network ports are associated with particular RPC services.

Security significance

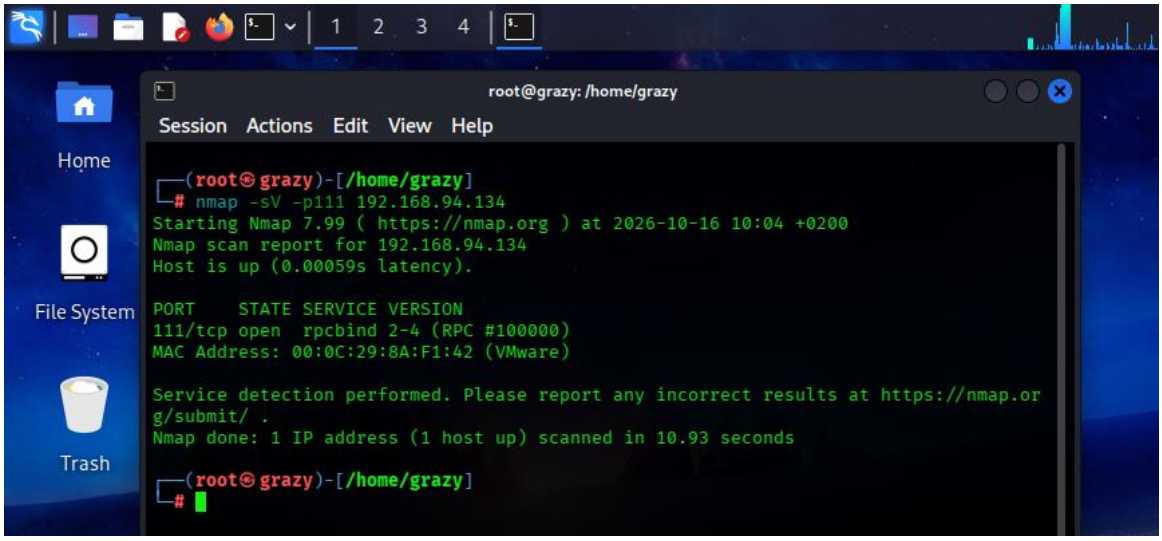
The presence of rpcbind is important during reconnaissance because it indicates that RPC functionality is exposed by the target.

From a security perspective, exposed services should be reviewed to determine whether they are necessary and whether access to them is appropriately restricted.

An exposed RPC-related service can also provide information about the services operating on the system. Therefore, discovering rpcbind gives the security analyst another area that may require investigation.

Importantly, the Nmap result alone does not prove that rpcbind is vulnerable. It only confirms that the service is accessible and identifies it.

Practical verification



The result confirms that RPC functionality is exposed on the target.

The key finding is: **Port 111 is open and provides the rpcbind service, contributing to the network attack surface of the target.**

10.4 Port 552002 - Status

Port 552002/tcp was also identified as open.

Nmap identified the service as:

- **status 1 (RPC #100024)**

This is an RPC-related service using a high-numbered TCP port.

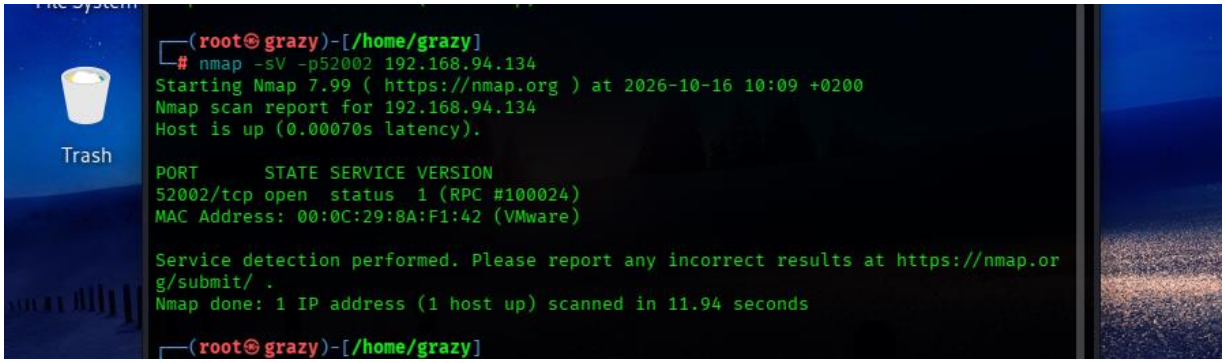
Security significance

High-numbered ports are commonly used by dynamically assigned services. Therefore, the discovery of port 552002 is important because it demonstrates that the target is providing additional network functionality beyond the commonly known services such as SSH and HTTP.

The service identification performed by Nmap is particularly useful here. Without service detection, an analyst would simply know that TCP port 552002 was open. With -sV, Nmap was able to associate the port with the status service.

This demonstrates the value of service and version detection during reconnaissance.

Practical verification



The result confirms that an additional RPC-related service is accessible on the target.

The key finding is: Port 552002 is open and provides the status RPC service, adding another exposed service to the target's attack surface.

10.5 Analysis of the Closed Ports

Nmap reported: **65,531 closed TCP ports**

This is important because the scan covered all 65,535 TCP ports.

Only four ports were identified as open:

- 22
- 80
- 111
- 552002

The remaining ports were reported as closed.

Security significance

A closed port generally indicates that there is no service accepting connections on that port at the time of the scan.

From an attack-surface perspective, this means that the target's externally accessible TCP services are concentrated around the four identified ports.

This helps the security professional focus subsequent analysis on services that are actually exposed rather than spending unnecessary time investigating ports where no accessible service was detected.

10.6 Overall Attack Surface Analysis

The Nmap scan revealed that the DC-1 system exposes several different types of network services.

The services can be grouped into three main categories:

- **Remote administration**

Port 22 provides SSH access.

- **Web services**

Port 80 provides an Apache HTTP service.

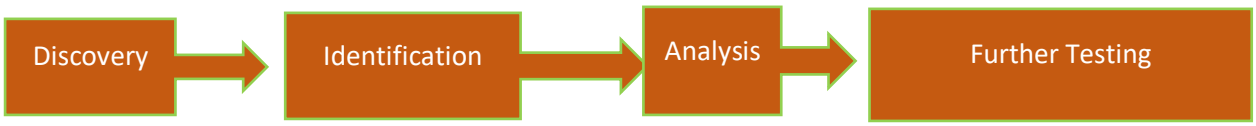
- **RPC services**

Ports 111 and 59368 provide RPC-related functionality.

This demonstrates an important cybersecurity concept: **every exposed service represents part of the attack surface that must be understood and appropriately secured.**

However, the presence of an open port should not automatically be interpreted as a vulnerability.

The correct security assessment process is:



Nmap performs the initial discovery and identification stages. Additional tools and manual analysis are then required to determine whether any actual security weaknesses exist.

10.7 Relationship Between Nmap and the Next Security Tools

The Nmap results directly influenced the next stages of the practical.

The discovery of port 80 was particularly important because it indicated the presence of a web server.

This allowed the assessment to continue with:



- Nmap identified the web service.
- Nikto was then used to examine the web server for interesting files, technologies, and common configuration issues.
- Firefox was used to access the web application.
- Burp Suite was then used to intercept and analyse HTTP requests generated by the browser.

This demonstrates how cybersecurity tools complement one another. One tool provides information that helps determine what should be examined next.

11. NIKTO WEB SERVER ASSESSMENT

Nikto is a web server assessment tool designed to examine web servers and identify potentially interesting security-related information.

While Nmap is primarily concerned with identifying hosts, ports, and network services, Nikto focuses specifically on web servers. This makes it useful after Nmap has identified an HTTP or HTTPS service.

In this practical, Nmap identified:

Port 80/tcp - Apache httpd 2.2.22 (Debian)

The discovery of an HTTP service provided a reason to perform further web-server analysis.

Nikto can examine a web server for information such as:

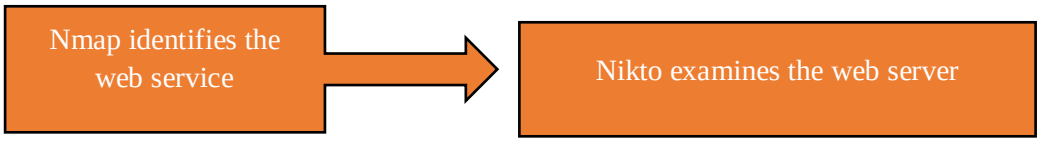
- Web server technology
- Server configuration information
- Interesting files and directories
- Commonly exposed installation files
- Potentially sensitive resources
- HTTP security headers
- Web technologies used by the application
- Other potentially interesting web-server observations

Nikto should be considered an assessment and information-gathering tool. A finding reported by Nikto does not automatically mean that the target has a confirmed vulnerability. Each finding requires further analysis to determine its actual security impact.

11.1 Purpose of Nikto in This Practical

The purpose of using Nikto was to obtain more detailed information about the web service discovered by Nmap.

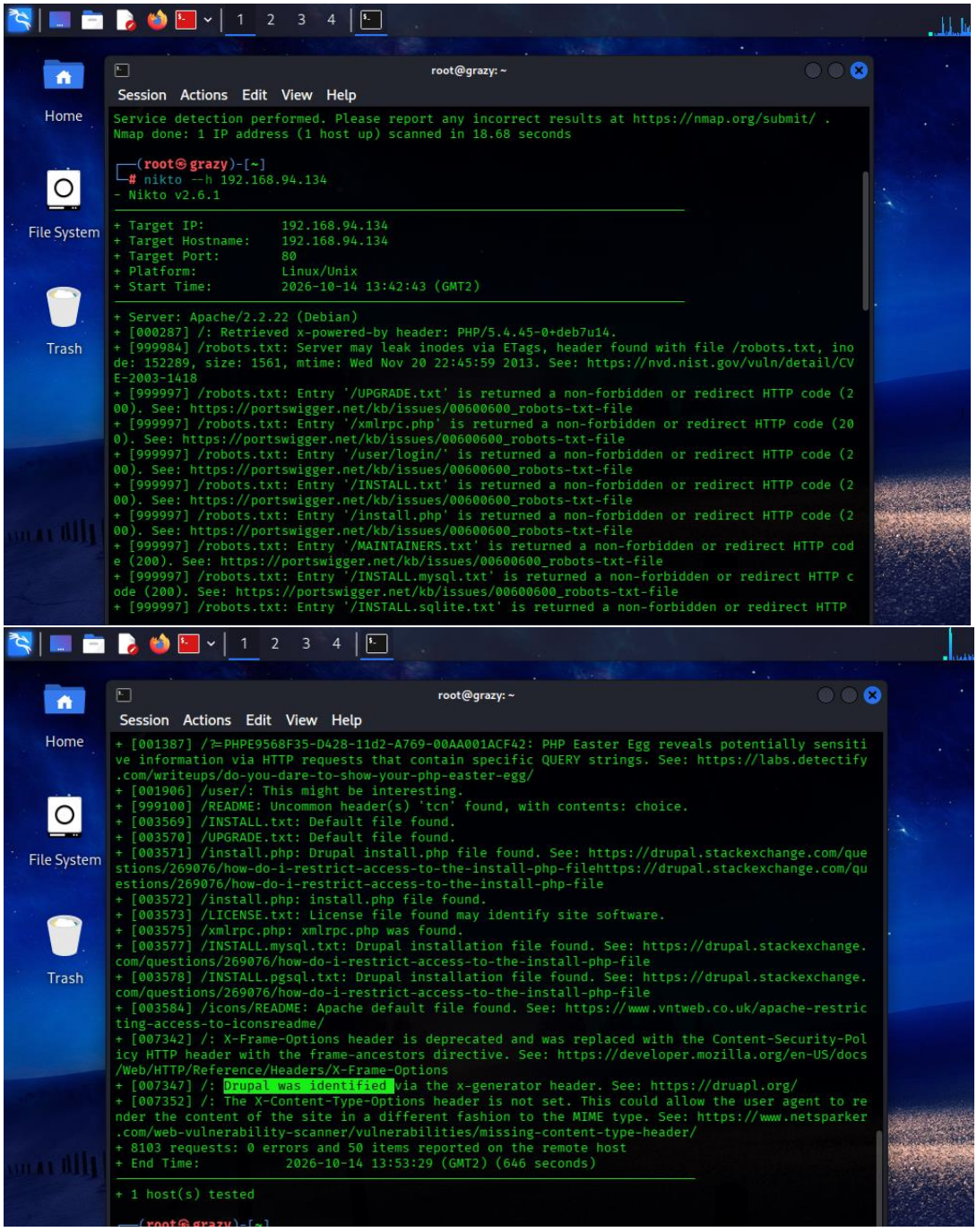
The relationship between the two tools can be understood as:



This demonstrates how different cybersecurity tools can be combined during an assessment.

Nmap established that port 80 was running Apache, while Nikto could then provide additional information about the web server and resources available through it.

11.2 Practical Nikto Scan



12. ANALYSIS OF NIKTO RESULTS

The Nikto assessment provided additional information about the web server that was not available from the initial Nmap scan.

The results identified the web server as Apache and also provided information suggesting that the website was using Drupal 7.

This is significant because identifying the underlying web technology helps a security professional understand what type of application is running on the target.

12.1 Apache Web Server

The Nikto results identified the web server as Apache.

Apache is a widely used web-server platform responsible for accepting HTTP requests and returning web content to clients.

In this practical, Apache is running on port 80, which was previously identified by Nmap.
This provides confirmation between the two assessment stages:

- **Nmap:** Identified Apache on port 80.
- **Nikto:** Performed additional analysis of the Apache web server.

The agreement between the tools increases confidence in the service identification.

12.2 Drupal 7 Identification

The Nikto output also indicated the presence of Drupal 7.
Drupal is a content management system used to create and manage websites and web applications.
From a security-assessment perspective, identifying the application framework or content management system is important because different technologies have different:

- Configurations
- Components
- Authentication mechanisms
- File structures
- Security considerations
- Update and maintenance requirements

However, identifying Drupal 7 alone does not prove that the application is vulnerable.
It simply provides useful technology information that can guide further authorized assessment.

12.3 Interesting Installation Files

Nikto identified several potentially interesting resources, including:

- **/INSTALL.txt**
- **/INSTALL.mysql.txt**
- **/install.php**

These files are significant because installation-related resources can sometimes reveal information about how an application was configured or deployed.
For example, an installation document may contain information about:

- Application setup
- Database configuration requirements
- Installation procedures
- Directory structures
- Configuration requirements

An exposed installation-related resource may therefore provide useful information to a security assessor.
The presence of such files does not automatically mean that sensitive information is exposed. Their security significance depends on what information can actually be accessed through them.

12.4 XML-RPC Resource

Nikto also identified: **/xmlrpc.php**
XML-RPC is a mechanism that allows applications to communicate using remote procedure calls through XML-based requests.
The identification of an XML-RPC endpoint is relevant because it indicates that the web application provides an additional communication interface.
From a security perspective, exposed application endpoints should be reviewed to determine:

- Why the endpoint is enabled
- What functionality it provides
- Whether authentication is required
- Whether access is appropriately restricted
- Whether the application is configured securely

Again, the existence of /xmlrpc.php alone does not prove a vulnerability.

12.5 Missing X-Frame-Options Header

Nikto reported that the web server was missing the:

- X-Frame-Options
- HTTP security header

The X-Frame-Options header can be used by a web application to control whether its pages can be displayed inside frames or iframes on another website.
This type of control can help reduce certain forms of UI redressing, commonly associated with clickjacking.
The absence of the header therefore represents a security-hardening observation.
It does not necessarily mean that the application is immediately exploitable. The actual risk depends on how the application behaves and whether sensitive actions can be performed through framed content.

12.6 Missing X-Content-Type-Options Header

Nikto also identified the absence of:
X-Content-Type-Options
This security header can be used to instruct browsers not to perform certain MIME-type sniffing behaviour.
Security headers are an important part of web application hardening because they provide additional controls over how browsers handle content.
The absence of this header does not automatically establish a critical vulnerability. Instead, it indicates that an additional browser-side security control is not being used.

12.7 Summary of Nikto Findings

The important Nikto observations can be summarized as follows:

Finding	Security significance
Apache web server	Identifies the web-server technology
Drupal 7	Identifies the web application technology
/INSTALL.txt	Installation-related resource exposed
/INSTALL.mysql.txt	Database installation-related resource
/install.php	Installation-related PHP resource
/xmlrpc.php	XML-RPC endpoint identified
Missing X-Frame-Options	Web security hardening issue
Missing X-Content-Type-Options	Browser security hardening issue

These findings demonstrate that web-server assessment can reveal information beyond simply identifying that port 80 is open.

13. WEB APPLICATION ACCESS USING FIREFOX

After identifying the HTTP service, the web application was accessed through Firefox.

A web browser is useful during a security assessment because it allows the tester to observe the application from the perspective of a normal user.

In this practical, Firefox was used to access: http://192.168.94.134

The page displayed the DC-1 web application.

This confirmed that the HTTP service discovered by Nmap was not only accepting network connections but was also delivering web content to a browser.

13.1 Importance of Browser-Based Testing

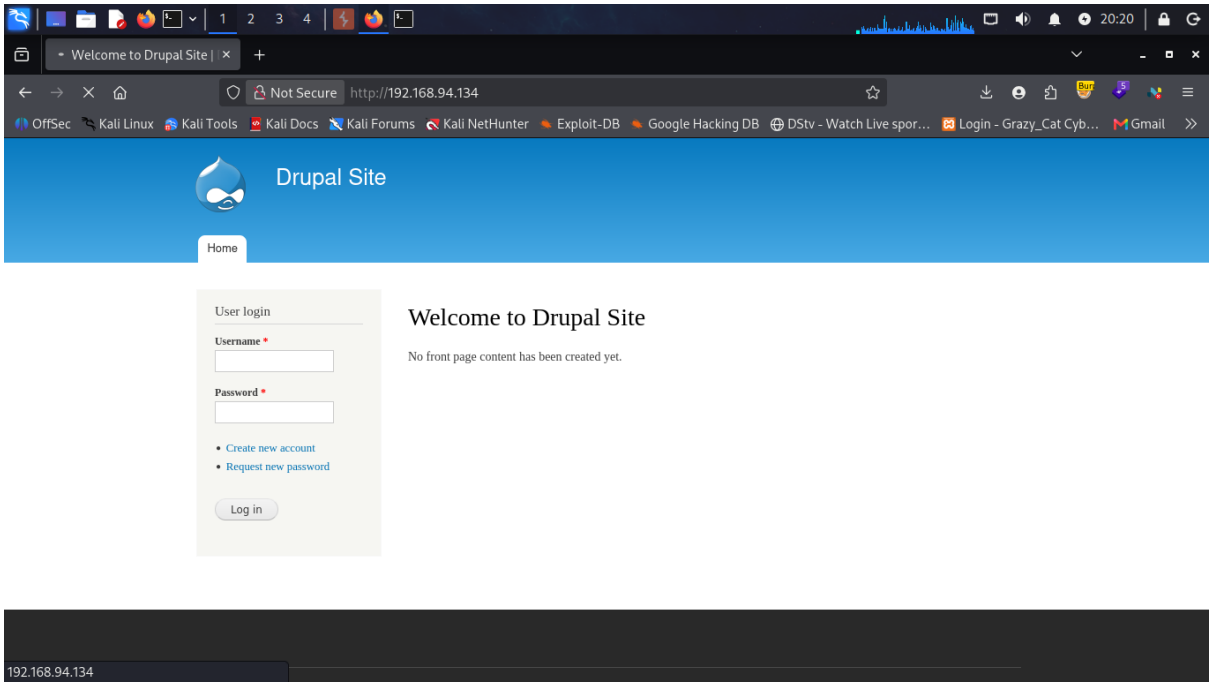
Browser-based testing provides information that a network scanner cannot provide by itself.

For example, a browser allows the tester to observe:

- Page structure
- Login forms
- Links
- Images
- Cookies
- Application responses
- HTTP behaviour
- User-facing functionality

This creates a transition from network reconnaissance to application-level analysis.

13.2 Practical Evidence



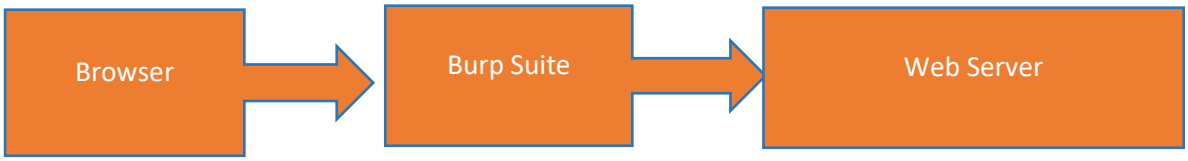
The screenshot demonstrates that the web service identified during Nmap scanning is accessible through HTTP.

14. BURP SUITE

Burp Suite is a web application security testing platform used to inspect and analyse HTTP and HTTPS communication.

One of its most important components is the Proxy, which can operate between the browser and the web application.

The basic communication model is:



Instead of the browser communicating directly with the web server, the request can pass through Burp Suite first.

This allows an authorized security tester to inspect the request before it reaches the target.

14.1 Purpose of Burp Suite

Burp Suite can provide visibility into information such as:

- HTTP methods
- URLs
- Hostnames
- Request headers
- Cookies
- User-Agent information
- Parameters
- HTTP responses

This is particularly useful because web applications depend heavily on HTTP requests and responses.

A browser may display only the final webpage, while Burp Suite allows the tester to examine the underlying communication that produced that webpage.

15. BURP SUITE PROXY CONFIGURATION

For this practical, Burp Suite was configured to work with Firefox as an HTTP proxy.

The purpose of the configuration was to allow HTTP requests generated by Firefox to pass through Burp Suite.

Once configured correctly, Burp Suite was able to capture requests generated when accessing the DC-1 web application.

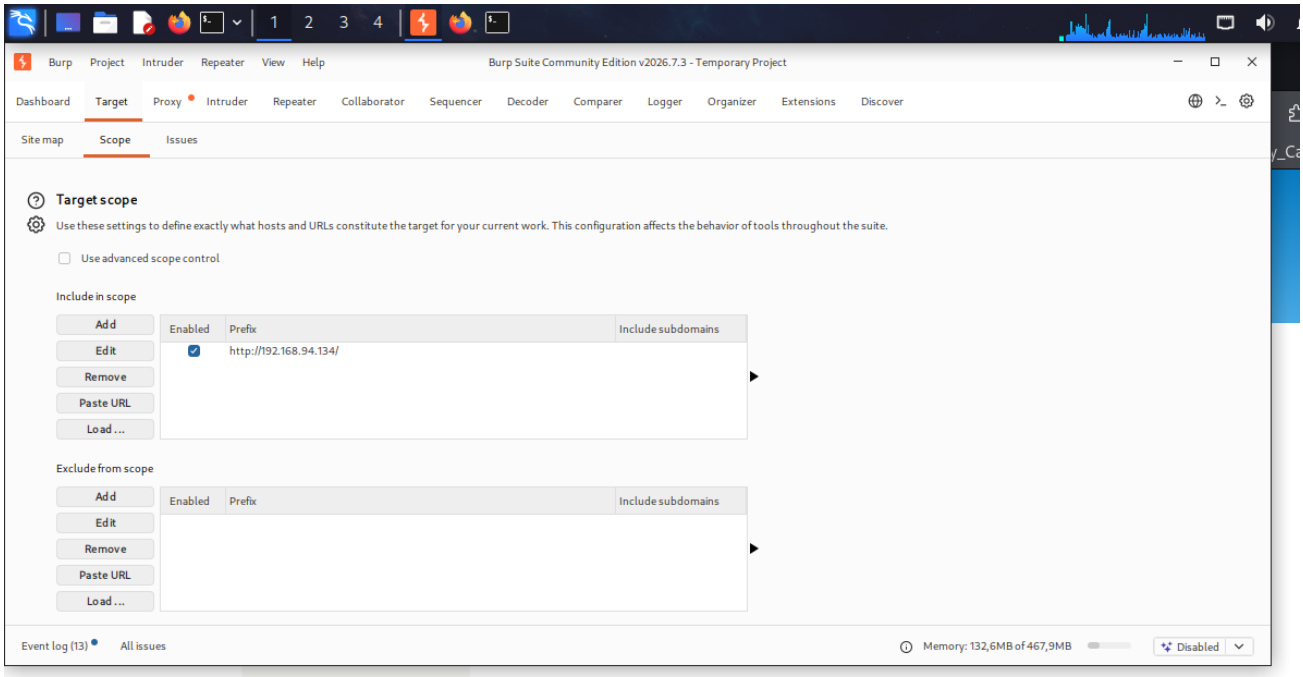
15.1 Target Scope

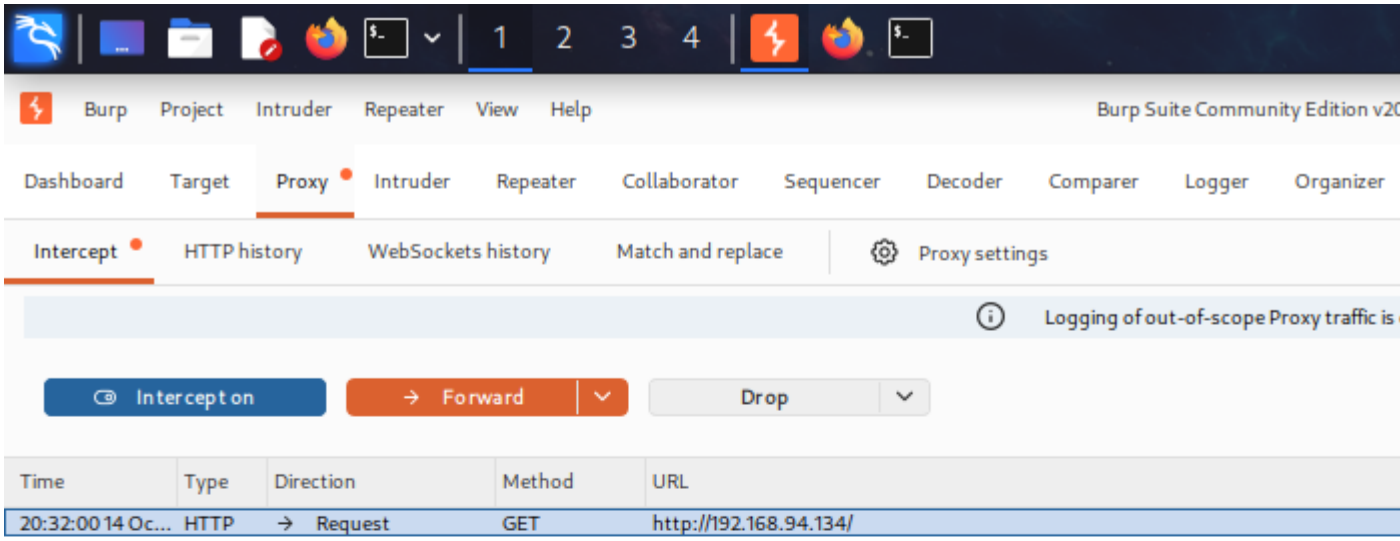
The DC-1 web application was added to the Burp Suite target scope: http://192.168.94.134/

Target scope is useful because it allows the tester to define which systems and applications are part of the authorized assessment.

In a real penetration-testing engagement, clearly defining the target scope is important for preventing accidental testing of systems outside the agreed authorization.

In this practical, the scope was limited to the intentionally vulnerable DC-1 machine.


Burp Project Intruder Repeater View Help
Burp Suite Community Edition v2026.7.3 - Temporary Project
Dashboard Target Proxy Intruder Repeater Collaborator Sequencer Decoder Comparer Logger Organizer Extensions Discover
Site map Scope Issues
Target scope
Use these settings to define exactly what hosts and URLs constitute the target for your current work. This configuration affects the behavior of tools throughout the suite.
Use advanced scope control
Include in scope
Add
Edit
Remove
Paste URL
Load ...
Enabled
Prefix
Include subdomains
http://192.168.94.134/
Exclude from scope
Add
Edit
Remove
Paste URL
Load ...
Enabled
Prefix
Include subdomains
Event log (13)
All issues
Memory: 132,6MB of 467,9MB
Disabled



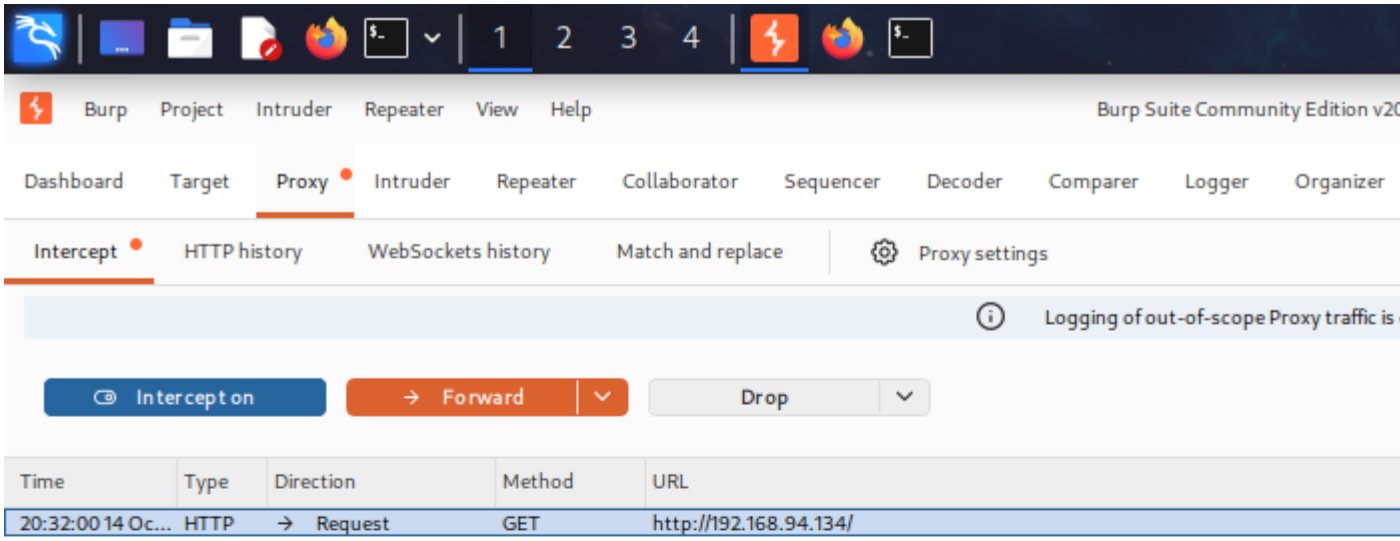
Request

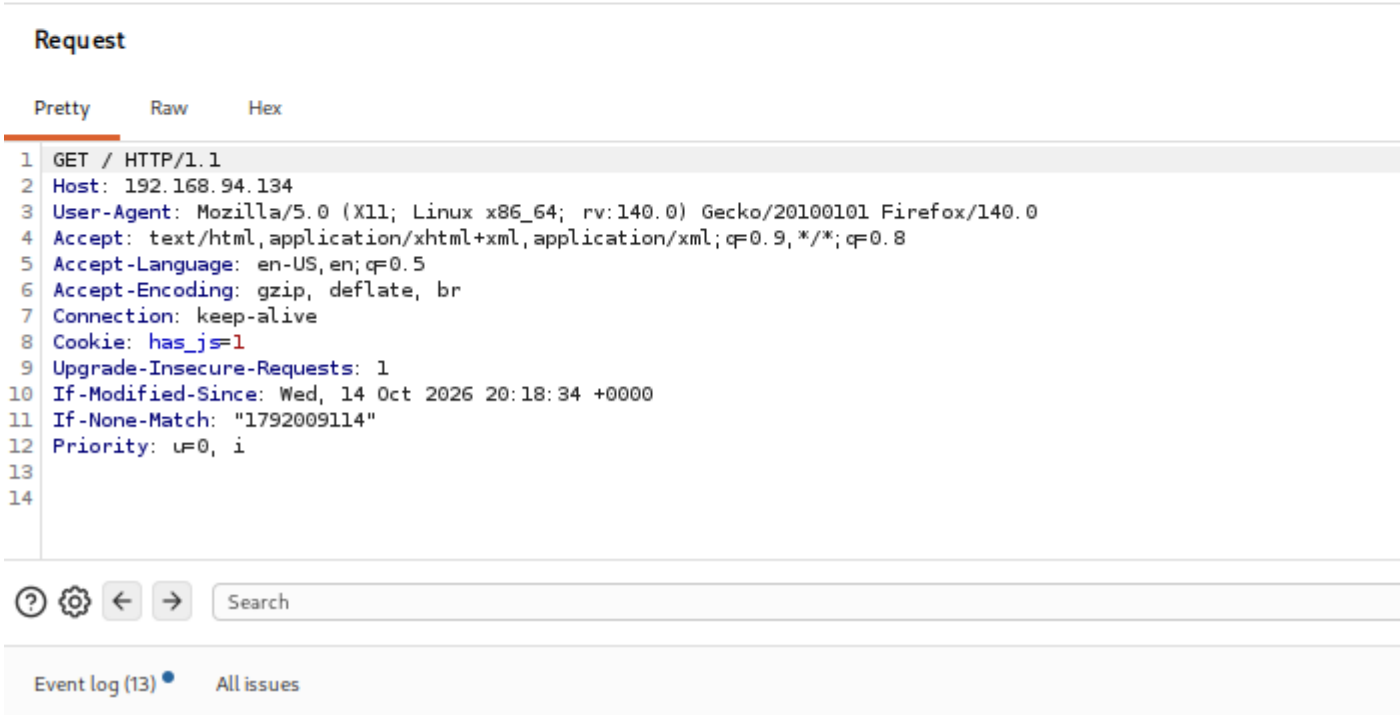
Pretty Raw Hex

```
1  GET / HTTP/1.1
2  Host: 192.168.94.134
3  User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0
4  Accept: text/html,application/xhtml+xml,application/xml;q=0.9,*/*;q=0.8
5  Accept-Language: en-US,en;q=0.5
6  Accept-Encoding: gzip, deflate, br
7  Connection: keep-alive
8  Cookie: has_js=1
9  Upgrade-Insecure-Requests: 1
10 If-Modified-Since: Wed, 14 Oct 2026 20:18:34 +0000
11 If-None-Match: "1792009114"
12 Priority: u=0, i
13
14
```

Search

Event log (13) All issues





16. HTTP REQUEST ANALYSIS USING BURP SUITE

After configuring Burp Suite, an HTTP request generated by Firefox was captured.

The intercepted request demonstrated how the browser communicates with the web server.

The request contained information such as the HTTP method, requested resource, host, browser information, and cookies.

One of the captured requests used:

GET

The GET method is commonly used to request a resource from a web server.

For example, a browser may request an image, webpage, stylesheet, or other resource using a GET request.

16.1 HTTP Request Method

The captured request used the:

- GET
- HTTP method.

The GET method is normally used when a client wants to retrieve a resource from a server.

This is different from methods such as POST, which are commonly used when submitting data to a server.

Understanding HTTP methods is important during web application security testing because different methods may perform different functions within an application.

16.2 Requested Resource

The captured traffic included requests for resources belonging to the DC-1 web application.

This demonstrates that loading a webpage does not necessarily generate only one HTTP request.

A browser may request:

- HTML documents
- Images
- CSS files
- JavaScript
- Fonts
- Other application resources

Burp Suite makes these individual HTTP communications visible to the security tester.

16.3 Host Header

The captured request contained the target host information associated with: 192.168.94.134

The Host header allows the web server to determine which host or website the client is attempting to access.

This is particularly important on systems hosting multiple websites or virtual hosts.

16.4 User-Agent Header

The captured request also contained browser identification information through the User-Agent header.

The User-Agent allows the server to receive information about the software making the request, such as the browser and operating system environment.

From a security perspective, HTTP headers provide useful information about how clients interact with web applications.

16.5 Cookies

The captured HTTP traffic also contained a cookie:

has_js=1

Cookies are pieces of information stored by the browser and sent with requests to websites.

Web applications can use cookies for purposes such as:

- Maintaining sessions
- Storing preferences
- Tracking application state
- Identifying a browser

In this case, the captured has_js=1 value provides an example of information transmitted by the browser as part of its interaction with the web application.

17. IMPORTANCE OF HTTP TRAFFIC ANALYSIS

The Burp Suite exercise demonstrated that web application security testing requires more than simply viewing a webpage.

The browser presents the application from a user perspective, while Burp Suite provides visibility into the underlying HTTP communication.

This allows a security tester to understand:



This is useful for identifying how an application handles:

- Requests
- Parameters
- Cookies
- Headers
- Authentication
- Sessions
- Application resources

Therefore, Burp Suite provides a deeper level of visibility than a normal browser alone.

18. NETCAT

Netcat is a network utility that can establish TCP or UDP connections between systems.

It is commonly used for:

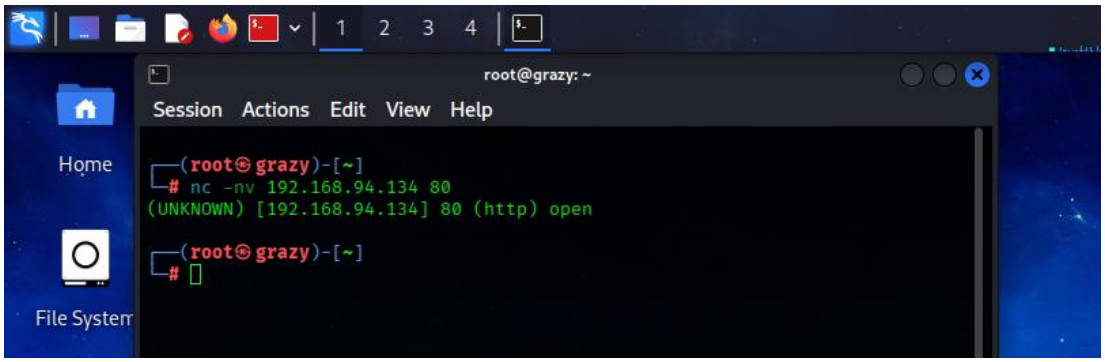
- Connectivity testing
- Troubleshooting
- Port testing
- Sending data
- Receiving data
- Testing network services

Unlike Nmap, which is primarily designed for scanning and service discovery, Netcat can be used to interact directly with a network service.

This makes it useful for verifying whether a specific port is reachable.

18.1 Netcat and Port 80

Because Nmap identified port 80 as open, Netcat can be used to test connectivity to that service.

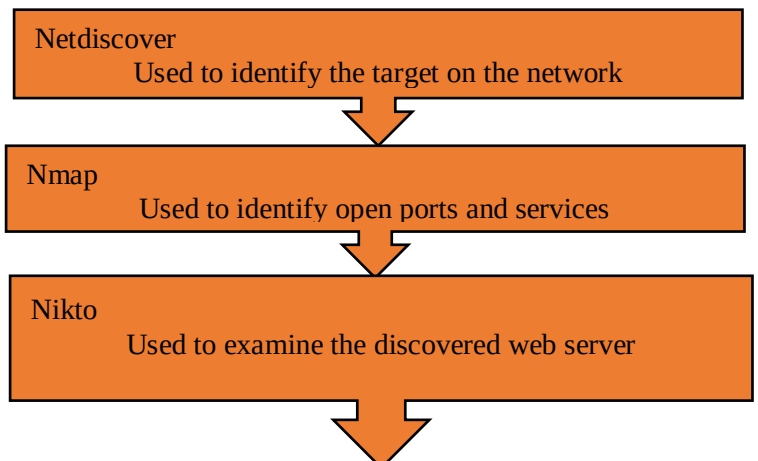


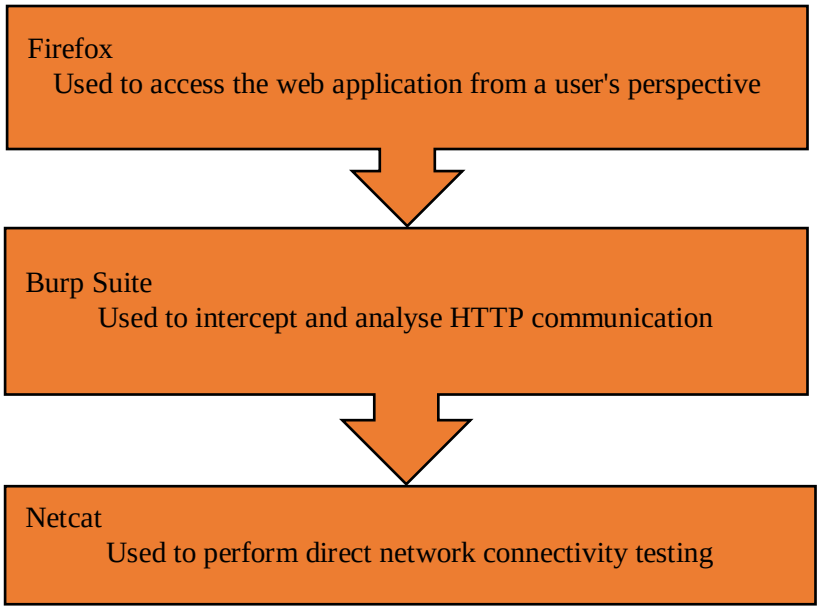
The screenshot provides practical evidence that the identified HTTP port can be reached from the Kali machine.

19. RELATIONSHIP BETWEEN THE SECURITY TOOLS

The practical demonstrated that each security tool provides a different perspective of the same target.

The tools can be viewed as a progressive assessment workflow:





This workflow demonstrates the importance of combining tools during cybersecurity assessments.

20. COMPARISON OF SECURITY TOOLS

Tool	Main purpose	Information obtained
Netdiscover	Network discovery	Hosts and IP addresses
Nmap	Port and service scanning	Open ports and services
Nikto	Web-server assessment	Web technologies and potential issues
Firefox	Web application access	User-facing application behavior
Burp Suite	HTTP analysis	Requests, responses, headers and cookies
Netcat	Network connectivity	Direct service connectivity

Each tool therefore contributes different information to the assessment

21. SECURITY OBSERVATIONS

The practical produced several important security observations.

First, the target exposed multiple network services. This demonstrates that the security of a system depends not only on the operating system itself but also on the services running on it.

Second, the HTTP service exposed a web application that could be accessed through a browser.

Third, Nikto identified additional web-server information and security-hardening observations.

Fourth, Burp Suite demonstrated that web requests contain important information such as HTTP methods, headers and cookies that may not be visible from the normal browser interface.

These observations demonstrate why layered security assessment is important.

22. CHALLENGES AND OBSERVATIONS

During the practical, one of the important considerations was ensuring that the correct target IP address was used throughout the assessment.

The target identified during discovery was: 192.168.94.134

Using the correct IP address was essential because Nmap, Nikto, Firefox, Burp Suite, and Netcat all needed to communicate with the same target.

Another important observation was that different tools provided different levels of information.

For example, Nmap identified the HTTP service, but Nikto provided additional information about the web server and application. Burp Suite then provided visibility into the HTTP communication itself.

This demonstrated that cybersecurity assessment is generally a multi-stage process rather than a single scan.

23. PRACTICAL RESULTS SUMMARY

Stage	Tool	Result
Network discovery	Netdiscover	Identified DC-1 at 192.168.94.134
Port scanning	Nmap	Identified four open TCP ports
Service identification	Nmap	Identified SSH, HTTP, rpcbind and status
Web assessment	Nikto	Identified Apache, Drupal and interesting resources
Web access	Firefox	Confirmed web application accessibility
HTTP analysis	Burp Suite	Intercepted and analysed HTTP requests
Connectivity testing	Netcat	Tested direct TCP connectivity

The results demonstrate the progression from network discovery to service identification and finally to web application analysis.

24. KEY LESSONS LEARNED

The practical provided several important cybersecurity lessons.

24.1 Reconnaissance is important

Before performing detailed security testing, it is important to understand the target environment.

24.2 Open ports reveal services

An open port can provide an entry point to a particular service and therefore contributes to the system's attack surface.

24.3 Service identification provides additional context

Knowing that port 80 is open is useful, but identifying Apache provides additional information that can guide further assessment.

24.4 Different tools provide different information

Nmap, Nikto, Burp Suite, Netcat, and Netdiscover perform different functions and therefore complement one another.

24.5 Web applications require application-level analysis

Identifying a web server is only the beginning. Understanding the application and its HTTP communication provides a deeper view of its security.

24.6 Security findings require interpretation

A tool reporting an observation does not automatically mean that a vulnerability exists. Findings must be analysed in context before determining their actual security impact.

25. OVERALL PRACTICAL WORKFLOW

The complete practical workflow was:

1. **Identify the target**

Netdiscover was used to identify the DC-1 IP address.

2. **Scan the target**

Nmap was used to scan all TCP ports and identify services.

3. **Analyse exposed services**

The open ports were analysed to understand the target's attack surface.

4. **Identify the web server**

Nmap **identified Apache running on port 80.**

5. **Assess the web server**

Nikto was used to identify web technologies, interesting resources, and security-header observations.

6. **Access the application**

Firefox was used to access the DC-1 web application.

7. **Intercept HTTP traffic**

Burp Suite was configured as a proxy to capture browser requests.

8. **Analyse HTTP communication**

Requests, headers, cookies, and resources were examined.

9. **Test network connectivity**

Netcat was used to verify connectivity to the HTTP service.

This workflow demonstrates a practical progression from reconnaissance to service analysis and web application assessment.

26. CONCLUSION

This practical provided an introduction to several commonly used cybersecurity tools and demonstrated how they can be combined during an authorized security assessment.

Netdiscover was used to identify the target, while Nmap provided detailed information about the target's exposed network services. The scan identified SSH on port 22, HTTP on port 80, rpcbind on port 111, and the RPC status service on port 59368.

The discovery of the HTTP service allowed the assessment to progress to web-server analysis using Nikto. The web application was then accessed through Firefox, while Burp Suite provided visibility into the HTTP requests generated by the browser.

Netcat provided an additional method of testing direct connectivity to a network service.

Overall, the practical demonstrated that effective cybersecurity assessment involves multiple stages and multiple tools. Each tool provides a different perspective, and combining their results allows a security professional to develop a clearer understanding of the target's attack surface.

Most importantly, the practical reinforced the principle that reconnaissance and enumeration should precede deeper security testing. By identifying what services are exposed before analysing them, security professionals can perform more structured, focused, and informed assessments within an authorized environment.